TEXORA · LECTURE SHEET

# Concepts of Colour and Colorants

*Light, the eye and the molecule: what colour is, how it is described and measured, and why conjugated molecules absorb visible light.*

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 1 · LEVEL 3 TERM 1

## 1. What colour is

Colour is not a substance but a sensation: light of certain wavelengths reaches the eye, is modified by selective absorption in the object, and is interpreted by the visual system. A red dyed fabric appears red because the dye absorbs most strongly in the green–blue region (about 500–560 nm) and reflects red light (620–700 nm). No useful discussion of colorants is possible without three ingredients — **illuminant** (D65 daylight, incandescent A, fluorescent F), **object** (the dyed textile, described by its reflectance spectrum), and **observer** (the CIE standard observers, 2° and 10°). Change any one and the perceived colour changes; two samples matching under one illuminant but not another show **metamerism**, the central problem of shade matching in mills.

## 2. Additive and subtractive mixing

**Additive mixing** adds lights: red + green + blue = white; it governs screens and stage lighting. **Subtractive mixing** removes wavelengths by absorption: cyan absorbs red, magenta absorbs green, yellow absorbs blue; cyan + magenta + yellow = black in principle. Dyeing and printing are subtractive — every dye added to a bath can only remove reflectance, never add it. That single fact explains why dye mixtures always become duller (lower chroma) as components are added, and why a bright shade can never be recovered by adding a fourth dye to a dull mixture.

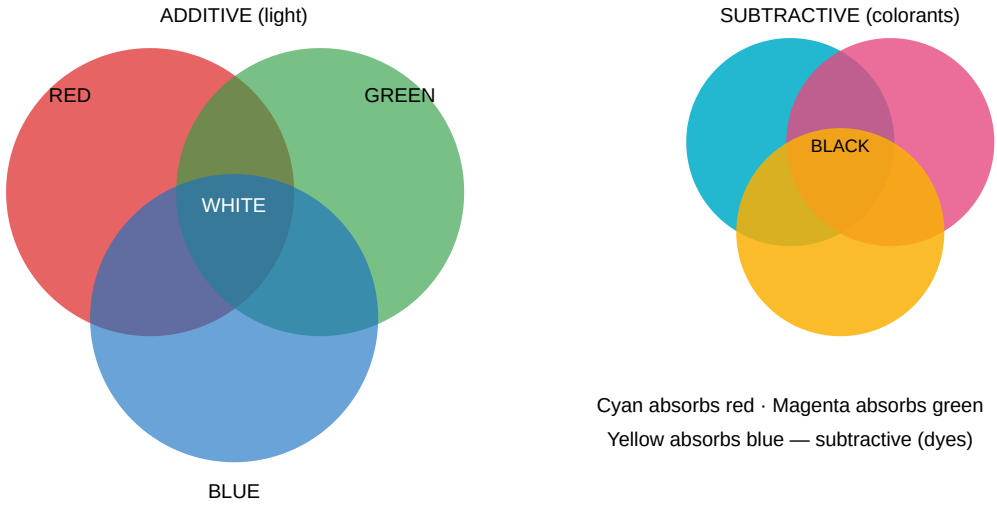



Fig. 1 — Additive mixing of lights (left) and subtractive mixing of colorants (right).

## 3. Hue, value and chroma — describing a colour

The Munsell system orders colour by **hue** (the family: red, yellow-red, yellow…), **value** (lightness, 0 black to 10 white) and **chroma** (saturation or purity, 0 at grey). The CIE system makes colour measurable: from a reflectance curve, tristimulus values X, Y, Z are computed for a chosen illuminant and observer, and converted to the uniform **CIELAB** space: L* (lightness 0–100), a* (green − to red +), b* (blue − to yellow +). Colour difference is the distance $\Delta E^*ab = \sqrt{[(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]}$; a ΔE of about 1 is near the limit of visual detection, and commercial pass/fail tolerances are commonly ΔE CMC(2:1) ≤ 1.0. Topic 11 shows how the reflectance and absorbance behind these numbers are measured.

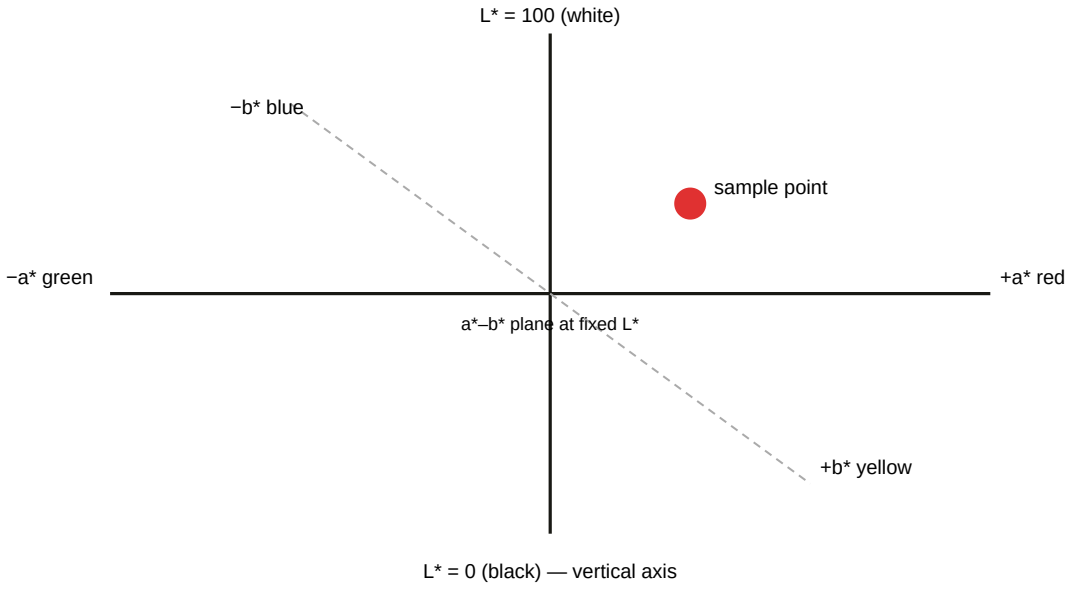



Fig. 2 — Sketch of the CIELAB colour space: L* vertical, a* and b* in the horizontal plane.

## 4. Why molecules are coloured — chromophores and auxochromes

A molecule is coloured when it absorbs visible light (400–700 nm), which requires an electronic transition whose energy gap falls in that range. Extended **conjugated systems** (alternating double bonds) shrink the HOMO–LUMO gap into the visible. The absorbing group is the **chromophore**: azo –N=N– (the largest dye class), carbonyl C=O in anthraquinones, nitro, methine and phthalocyanine macrocycles. **Auxochromes** (–OH, $–NH_2$, $–SO_3H$, –COOH) do not create colour alone but modify it: electron-donating auxochromes shift absorption to longer wavelength (**bathochromic shift**, towards red) and increase intensity (**hyperchromic effect**); protonation or deprotonation of auxochromes is why many dyes change shade with pH. A shift to shorter wavelength is **hypsochromic**; a fall in intensity is **hypochromic**. Sulphonate auxochromes also confer the water solubility most dye classes depend on.

## 5. Colorant classification — the working framework

| BASIS | CLASSES / EXAMPLES |
|---|---|
| By chemical structure | Azo, anthraquinone, phthalocyanine, triphenodioxazine, formazan, indigoid, sulphur |
| By application (Colour Index generic name) | Acid, basic (cationic), direct, disperse, mordant, reactive, sulphur, vat, pigment, solvent |
| By ionic character in solution | Anionic (acid, direct, reactive, most vat leuco forms), cationic (basic), non-ionic (disperse, pigments) |

The **Colour Index (CI)** gives every commercial colorant a generic name (e.g. CI Reactive Blue 19) and a constitution number (CI 61200) identifying its chemical structure when disclosed. Trade names vary by supplier; the CI name is the only stable identity, and mill recipes should always record it.

## 6. Colour fastness — the quality dimension

A colorant is commercially useful only if the colour survives washing, rubbing, light, perspiration and ironing. Fastness is graded 1 (very poor) to 5 (excellent) against grey scales for change of shade and staining (Experiment 5 of TE-3106 works the tests). Fastness is a property of the dye–fibre *system*, not the dye alone: the same dye shows different wash fastness on cotton and on nylon because the binding forces differ. The binding forces themselves are the subject of Topic 2.

## 7. Glossary — key terms

- **Metamerism:** a pair matching under one illuminant and mismatching under another.

- **Chromophore:** the unsaturated group responsible for visible absorption.
- **Bathochromic shift:** absorption shift to longer wavelength (towards red).
- **CIELAB:** CIE L*a*b* uniform colour space used for instrumental shade sorting.
- **Chroma:** saturation — the departure of a colour from a grey of the same lightness.

## 8. Key points

- Colour = illuminant × object × observer; dyeing is subtractive mixing.
- Conjugation creates colour; auxochromes tune shade, intensity and solubility.
- CIELAB and ΔE turn shade approval into a measurement, not an argument.
- Always specify colorants by CI generic name; judge them as dye–fibre systems including fastness.

## 9. Summary

Colour begins as physics — selective absorption of visible light — and becomes chemistry through conjugated chromophores assisted by auxochromes. Mills work with two maps of the same territory: the descriptive Munsell order and the measurable CIELAB space, joined by the Colour Index naming system. Every later topic in this course — dye selection, recipe calculation, fastness testing, spectrophotometry — is an application of these foundations.

## 10. Practice

1. A sample matches a standard under D65 but looks greener under shop fluorescent light. Name the phenomenon and state two practical defences in a dyehouse.
2. Sample A: L* 52, a* 38, b* 12. Sample B: L* 50, a* 40, b* 10. Calculate ΔE*ab and comment on acceptability against a tolerance of 1.0.
3. Explain, using HOMO–LUMO ideas, why lengthening a conjugated system shifts absorption bathochromically.
4. Why does adding a fourth dye to correct a dull shade almost always make it duller still?

### Answer key

1. Metamerism. Defences: match under at least two illuminants (D65 and the buyer's store light, e.g. TL84/F11); prefer dye combinations with similar reflectance-curve shapes to the standard (low-metamerism recipes), verified instrumentally.
2. ΔL* = −2, Δa* = +2, Δb* = −2; ΔE = √(4+4+4) = √12 = 3.46. Far above 1.0 — reject; the batch is darker, redder and less yellow.

3. More conjugated atoms share the π electrons over a longer system, narrowing the HOMO–LUMO gap; a smaller energy gap is bridged by a lower-energy (longer-wavelength) photon, so absorption moves towards the red.
4. Dyeing is subtractive: each added dye absorbs (removes) another band of reflectance. Total reflected light and therefore chroma can only fall; brightness lost to absorption cannot be added back.

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